

A Technical Introduction to Image super-resolution

Section 1

Known defects in a given imaging situation, such as defocus or aberrations, can sometimes be mitigated in whole or in part by suitable spatial-frequency filtering of even a single image. Such procedures all stay within the diffraction-mandated passband, and do not extend it.

Section 2

Sub-pixel image localization The location of a single source can be determined by computing the "center of gravity" (centroid) of the light distribution extending over several adjacent pixels (see figure on the left). Provided that there is enough light, this can be achieved with arbitrary precision, very much better than pixel width of the detecting apparatus and the resolution limit for the decision of whether the source is single or double. This technique, which requires the presupposition that all the light comes from a single source, is at the basis of what has become known as super-resolution microscopy, e.g. stochastic optical reconstruction microscopy (STORM), where fluorescent probes attached to molecules give nanoscale distance information. It is also the mechanism underlying visual hyperacuity.

Section 3

Other related work Curtis, Craig H.; Milster, Tom D. (October 1992). "Analysis of Superresolution in Magneto-Optic Data Storage Devices". *Applied Optics*. 31 (29): 6272–6279. Bibcode:1992ApOpt..31.6272M. doi:10.1364/AO.31.006272. PMID 20733840. Zalevsky, Z.; Mendlovic, D. (2003). *Optical Superresolution*. Springer. ISBN 978-0-387-00591-1. Caron, J.N. (September 2004). "Rapid supersampling of multiframe sequences by use of blind deconvolution". *Optics Letters*. 29 (17): 1986–1988. Bibcode:2004OptL...29.1986C. doi:10.1364/OL.29.001986. PMID 15455755. Clement, G.T.; Huttunen, J.; Hynynen, K. (2005). "Superresolution ultrasound imaging using back-projected reconstruction". *Journal of the Acoustical Society of America*. 118 (6): 3953–3960. Bibcode:2005ASAJ..118.3953C. doi:10.1121/1.2109167. PMID 16419839. Geisler, W.S.; Perry, J.S. (2011). "Statistics for optimal point prediction in natural images". *Journal of Vision*. 11 (12): 14. doi:10.1167/11.12.14. PMC 5144165. PMID 22011382. Cheung, V.; Frey, B. J.; Jojic, N. (20–25 June 2005). *Video epitomes* (PDF). Conference on Computer Vision and Pattern Recognition (CVPR). Vol. 1. pp. 42–49. doi:10.1109/CVPR.2005.366. Bertero, M.; Boccacci, P. (October 2003). "Super-resolution in computational imaging". *Micron*. 34 (6–7): 265–273. doi:10.1016/s0968-4328(03)00051-9. PMID 12932769. Borman, S.; Stevenson, R. (1998). "Spatial Resolution

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Section 4

Some object features, though beyond the diffraction limit, may be known to be associated with other object features that are within the limits and hence contained in the image. Then conclusions can be drawn, using statistical methods, from the available image data about the presence of the full object. The classical example is Toraldo di Francia's proposition of judging whether an image is that of a single or double star by determining whether its width exceeds the spread from a single star. This can be achieved at separations well below the classical resolution bounds, and requires the prior limitation to the choice "single or double?" The approach can take the form of extrapolating the image in the frequency domain, by assuming that the object is an analytic function, and that we can exactly know the function values in some interval. This method is severely limited by the ever-present noise in digital imaging systems, but it can work for radar, astronomy, microscopy or magnetic resonance imaging. More recently, a fast single image super-resolution algorithm based on a closed-form solution to $\ell_2 - \ell_2$ problems has been proposed and demonstrated to accelerate most of the existing Bayesian super-resolution methods significantly.

Section 5

Technical implementations There are many both single-frame and multiple-frame variants of SR. Multiple-frame SR uses the sub-pixel shifts between multiple low resolution images of the same scene. It creates an improved resolution image fusing information from all low resolution images, and the created higher resolution images are better descriptions of the scene. Single-frame SR methods attempt to magnify the image without producing blur. These methods use other parts of the low resolution images, or other unrelated images, to

guess what the high-resolution image should look like. Algorithms can also be divided by their domain: frequency or space domain. Originally, super-resolution methods worked well only on grayscale images, but researchers have found methods to adapt them to color camera images. Recently, the use of super-resolution for 3D data has also been shown.